

Debugging shapes and gradients

Project: Image-pattern Network Investigation

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Initialisation and reproducibility

Without opening the old lesson, explain one key idea from Initialisation and reproducibility. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Training and validation curves

Without opening the old lesson, explain one key idea from Training and validation curves. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Training a small network

Without opening the old lesson, explain one key idea from Training a small network. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Expose a hidden bias error for Debugging shapes and gradients. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Debugging shapes and gradients. Proposed explanation: “Matching output shape guarantees the computation”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Debug a new gradient independently

Use $y = 3a + c$, target 2, loss $(y - 2)^2$, at $[a, c] = [0, 1]$. A program returns gradient $[-2, -6]$ in order $[a, c]$. Use symmetric step 0.1 to check each coordinate, then update both with the corrected gradient and rate 0.05. Report new parameters and loss.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Reject a shape mismatch

The layer expects W shape 3 by 2, b length 3 and x length 2. A submitted x has length 3. Should the last feature be silently ignored? State the failed contract and repair process.

My reasoning and evidence

6. W2 Expose a hidden bias error

Compute scores and activations for $x = [3, 1]$ with the lesson layer, then omit the third bias. Identify the first discrepancy and the resulting activation error.

My reasoning and evidence

7. W3 Check an update sign

For the diagnostic model at $[1, 0]$, gradient $[-4, -2]$ and rate 0.05, calculate the correct new parameters and loss. Compare with adding the gradient step.

My reasoning and evidence

8. W4 Debug a new gradient independently

Use $y = 3a + c$, target 2, loss $(y - 2)^2$, at $[a, c] = [0, 1]$. A program returns gradient $[-2, -6]$ in order $[a, c]$. Use symmetric step 0.1 to check each coordinate, then update both with the corrected gradient and rate 0.05. Report new parameters and loss.

My reasoning and evidence

9. W5 Keep independent regression checks

A program passes only the $x = [1, 2]$ final-activation test. Propose one intermediate-value check, one malformed-input check and one additional input that exposes the bias omission.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled debugging shapes and gradients evidence sheet to Image-pattern Network Investigation. Show how you can check dimension contracts before computing a layer. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Debugging shapes and gradients

Catch a gradient -order bug before the optimiser changes parameters.

Toy $y=2a+3c$, target 4, start $[a, c]=[1, 0]$. Loss $(y-4)^2$. Program claims gradient $[-12, -8]$ in order $[a, c]$.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Debugging shapes and gradients

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Compute correct gradient

$y = 2$, residual -2 : gradient $[-8, -12]$, not $[-12, -8]$.



2. Check coordinates

Step 0.1: a losses 3.24, 4.84 give -8 ; c losses 2.89, 5.29 give -12 .



3. Apply correct update

At rate 0.01: new $[a, c] = [1.08, 0.12]$; $y = 2.52$, loss 2.1904.

CHANGE ONE THING

The program changes its declared order to $[c, a]$ and returns $[-12, -8]$.

Explain the changed result using the matching lesson.

What should a debugging record pair with every gradient value? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

13. Project portfolio milestone

Project: Image-pattern Network Investigation

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Catch a gradient -order bug before the optimiser changes parameters.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? At rate 0.01: new [a, c]=[1.08, 0.12]; y = 2.52, loss 2.1904.

1. Compute correct gradient
2. Check coordinates
3. Apply correct update

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Apply correct update At rate 0.01: new [a, c]=[1.08, 0.12]; y = 2.52, loss 2.1904. Array shape and coordinate meaning both matter. A vector with two entries can have the right length but assign each derivative to the wrong parameter. Label the entries a and c, then perturb one coordinate at a time to check their slopes. Repairing the coordinate mapping is necessary before using the gradient to update the network.

Practice 2: explain the deciding evidence

Changed-case practice: The program changes its declared order to [c, a] and returns [-12,-8]. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future input.
2. That vector now matches the declared coordinate order.
3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: That vector now matches the declared coordinate order. Gradient values are meaningful only when their parameter names and array positions agree.

Repair a missing building block

Review Initialisation and reproducibility (ai-5/17). Goal: Trace how starting values affect a fixed training budget. Short practice: Does the matching final loss mean the parameter values are equal? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Training and validation curves (ai-5/15). Goal: Compute comparable training and validation losses at saved checkpoints. Short practice: Can the original epoch 3 training loss 1 justify claiming best generalisation? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Training a small network (ai-5/12). Goal: Trace repeated forward loss gradient and update steps. Short practice: Which parameter changed, and which part was not trained? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
